Peter Hou | Curriculum Vitae

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Profile

Microsoft — **Redmond, WA, USA**
Full Time Employee, Software Engineer of Agent365 Team — *Feb 2023 – Present*

Microsoft — **Suzhou, China**
Full Time Employee, Software Engineer of WebXT [eXperiences Team] — *Sep 2020 – Feb 2023*

Deep Motion — **Beijing, China**
Full Time Employee, Deep Learning R&D Engineer — *Nov 2018 – Aug 2020*

Microsoft Research Asia — **Beijing, China**
Intern, System Research Group — *July 2018 – Nov 2018*

Shanghai Jiao Tong University (SJTU) — **Shanghai, China**
Bachelor of Engineering, Outstanding Graduate (Top 3%) — *Sep 2014 – July 2018*

- School: Electronic Information and Electrical Engineering, Department: Electronic Engineering (EE)
- GPA of Upper Division Work: **3.83/4.3(89.59/100)**, Standing: **4/60**

Preprint

- **Bitan Hou**, Yujing Wang, Ming Zeng, Shan Jiang, Ole J. MengShoel, Yunhai Tong, Jing Bai. Customized Graph Embedding: Tailoring Embedding Vectors to different Applications. [arXiv]

Work & Research Experience

- **Enterprise AI Agents Control Plane: Microsoft Agent 365**
  - Built the Agent Overview experience from scratch in weeks, which became a Microsoft Ignite 2025 show case and served as the entry point for Microsoft Agent 365
  - Designed and implemented the whole infrastructure from backend to frontend, from API signature to modern React practice, from data management to page render performance
- **Copilot Search Experience in Microsoft Admin Center**
  - Delivered two high-impact features - Acronyms and Bookmarks - as core components of the Copilot Search enterprise experience
  - Scaled and deployed the features to U.S. Government cloud environments
  - Owned end-to-end development including implementation, telemetry/logging, flighting controls, metrics, and subscription eligibility validation, etc.
- **Full-Stack Engineer of Customer Self-Help System for Microsoft 365 Products**
  - Built a backend service from scratch using the latest .Net technology stack
  - Established an end-to-end code rollout process, including build pipeline, release pipeline, global deployments, Azure resource management, and system health monitoring
  - Integrated GPT into the customer self-Help system for enhanced user assistance
- **User Experience Enhancement of Microsoft Web Products (NTP, Bing HP, MSN)**
  - Owned the search experience for the NTP (**N**ew **T**ab **P**age) in Microsoft Edge, a high-traffic surface, including search box UI, search history management, and voice search, driving measurable improvements in user engagement and revenue
  - Improved News Feed and advertising experiences, contributing directly to revenue growth
  - Acted as DRI for Bing Homepage, ensuring high availability and quality for global users
  - Contributed to a major web framework migration from React to Web Components, significantly improving page performance and rendering efficiency
  - Expanded automated testing coverage for NTP, including unit tests, integration tests, and visual regression (parity) tests

- **User Nurturing Platform of Microsoft Edge NTP**
  - Independently rebuilt the user nurturing platform for NTP, which is a crucial way to promote new features to our users
  - Maintained the stability of the platform, and supported many coachmarks from global teams which brought millions of clicks/downloads of Microsoft products
  - Addressed the pain points of original promotion pipeline, and creatively leveraged the power of team WPO (**W**hole **P**age **O**ptimization) which can generate user's profile with Machine Learning algorithm
- **Data Analysis**
  - Conducted large-scale analysis of user telemetry to quantify the impact of new features
  - Drove data-informed product iteration by utilizing internal data platforms, becoming a key contributor to feature evaluation and decision-making.
- **Model Acceleration & Deployment (100k+ lines of C++)**
  - Had expertise in using various popular Deep Learning Edge Devices for practical applications, such as NVIDIA Xavier, TX2, Nano, HUAWEI Atlas200DK and TDA3x of Texas Instruments (TI)
  - Deployed 80+ models with expertise in the NVIDIA **TensorRT** platform for high-performance inference
  - Familiar with both ARM and x86 architectures, both CPU and GPU instructions for ML Deployment
- **Quantization**
  - Dived into QNNPACK, a Caffe2 8-bit quantization framework, and applied to algorithms within one month of its release from Facebook;
  - Widely used due to its highly efficient performance (**1/4 size, 5x speed, only 1% AP drop**)
  - Conducted the theoretical analysis of quantization in different tasks (Cls., Seg., Det., Depth) by using the newest INT8 Quantization feature in PyTorch1.3; Deployed 10+ INT8 models based on TorchScript
- **Deep Learning Framework Development**
  - Got familiar with Google XNNPACK, a highly optimized library of neural network inference operators
  - Developed a lite deep learning framework (inference only) for edge devices deployment which needs strong skill on Google FlatBuffers, memory management, topology optimization, etc.
- **Neural Architecture Search (NAS)**
  - Reproduced DARTS, Proxyless NAS (Song Han), Auto-DeepLab (Feifei Li) and Ramdom NAS
  - Extended NAS to **dense image prediction**
- **Training Efficiency**
  - Reduced the training time from 28 GPU-days to 4 GPU-days on GTX-1080Ti by using NVIDIA Data Loading Library (**DALI**) without accuracy reduction (Train ImageNet from scratch)
  - Used **Mixed Precision Training** based on *Tensor Cores* and introduced by Volta Generation of GPUs, to enlarge 8x through put and no accuracy reduction
- **Graph Embedding**
  - **Reproduced** papers related to graph embedding, such as DeepWalk, Node2Vec, and Plantoid
  - Proposed a novel semi-supervised approach, **Customized Graph Embedding**, which significantly improved the performance of clustering and representation
  - Completed a **first-author paper** in collaboration with MSRA, CMU, UIUC, PKU and SJTU
- **Face Recognition (CNN)**
  - Independently developed a face recognition system for a commercial applications, such as city security
  - **Excellent** performance in both face comparison(95.53% on YTF) and identity verification(99.95%)
- **Photonic Crystals**
  - Conducted research on photonic crystals; analyzed simulations using **MEEP** (an MIT electromagnetic simulation software); summarized findings in a report: *Dynamic control of optical pulse delay time*
  - Honored by Tsung-Dao Lee Chinese Research Program with the title "Distinguished Scholar"
  - Selected from the top 3% of applicants to this program supported by Tsung-Dao Lee
- **Misc.** (All of them are implemented for widely use within our company.)
  - Developed a python package for model conversion between DL frameworks (**10k+ lines of .py**)
  - Self-developed an OpenCV (GPU) package to get ride of redundant dependencies (**2k+ lines of .cu**)
  - Python Binding: Created **100+ bindings** of existing C++ code, using C++ code through python API
  - Developed a C++ library for Caffe parser using Google Protocol Buffers (**2k+ lines of .cpp**)
  - Developed a toolchain for CNN debugging with expertise in python features: hook and decorator

Honors & Awards During School Years

Outstanding Graduate of SJTU (Top 3%, performances over four years are considered) *2018*
Honorable Mention of Mathematical Contest in Modeling(MCM), America *2017*
Second Prize in National Undergraduate Electronics Design Contest(Shanghai) (Top 10%) *2017*
Tsung-Dao Lee Scholarship (Top 3%, sponsored by the recipient of Nobel Prize in Physics) *2017*
Ji Hanbing Alumnus Scholarship (Only 1 in my major, honoring academic excellence) *2017*
Liu Yongling Fellowship(Hong Kong) (Only 1 in my major, honoring academic excellence) *2017*
Academic Excellence Scholarship (Second Class) of SJTU (Top 3 in my major) *2017*
The Merit Student of SJTU (Only 1 in my major, comprehensive evaluation) *2016*
National Endeavor Fellowship of Shanghai Jiao Tong University (Top 1%, national level) *2016*
Third Prize in China Undergraduate Mathematical Contest in Modeling(Shanghai, China) *2016*
Academic Excellence Scholarship (Third Class) of Shanghai Jiao Tong University *2016*
Third Prize in Mettler Toledo Innovation Contest *2016*
Third Prize in China Undergraduate Physical Experiment Contest (SJTU) *2016*
Third Prize in Texas Instruments(TI) Cup Electronic Design Contest (SJTU) (Top 10) *2016*

Conferences, Short-Term Programs, Voluntary and Social Activities

Computing in the 21st Century Conferences & MSRA Faculty Submmit
Invited audience. One-on-one talk with Kai-Fu Lee and Harry Shum, respectively. 2018

Building Bridges Education Support Program, with Yale U(Organizer), Hong Kong U, Peking U
Team leader. Certificated by the Aixin Foundation Inc. of the United States. 2017

Tsinghua University(THU) Summer Camp: Nano-OptoElectronics Lab
Certificate as outstanding participant by Department of Electronic Engineering, THU 2017

Career & Leadership Development Program
Served as coach. Certified by China Soong Ching Ling Foundation, Liaison Department. 2015

Shanghai International Marathon
Volunteer. Served thousands of athletes and running enthusiasts from all over the world. 2014, 2025, 2016

Skills (Traditional engineering foundations evolving alongside modern AI-assisted workflows)

- **Programming**: Python, C++11/14, CUDA, Verilog/VHDL, C#, React (TS/JS, HTML, CSS), Neon
- **Dev. SKills**: .Net, Azure(CI/CD, cloud infra), Linux OS, Modern CMake, GDB, Git, Vim, Emacs
- **DL Frameworks**: PyTorch, Caffe, Caffe2; Familiar with Theano, Keras, Tensorflow, MxNet

Interests

- Reading, travel (Hawaii enthusiast), music, ultimate frisbee, American football (NFL), baseball, pickleball